Measurement of Absolute Double Differential Cross-Section in Invariant Mass and p_T Bins

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May 11, 2026

1 Input Files Used

Note: We do not use RS5a and RS5b files for this study.

ROOT files used:

- merged_RS57_Empty_2_1138_FinalData.root
- merged_RS57_LD2_3_1138_FinalData.root
- merged_RS57_LH2_1_1138_FinalData.root
- merged_RS59_Empty_2_466_FinalData.root
- merged_RS59_LD2_3_466_FinalData.root
- merged_RS59_LH2_1_465_FinalData.root
- merged_RS5a_Empty_2_1680_FinalData.root
- merged_RS5a_LD2_3_1689_FinalData.root
- merged_RS5a_LH2_1_1476_FinalData.root
- merged_RS5b_Empty_2_917_FinalData.root
- merged_RS5b_LD2_3_918_FinalData.root
- merged_RS5b_LH2_1_770_FinalData.root
- merged_RS62_Empty_2_1234_FinalData.root
- merged_RS62_LD2_3_1237_FinalData.root
- merged_RS62_LH2_1_1234_FinalData.root
- merged_RS67_Empty_2_14_FinalData.root
- merged_RS67_LD2_3_15_FinalData.root
- merged_RS67_LH2_1_5_FinalData.root
- merged_RS70_Empty_2_267_FinalData.root
- merged_RS70_LD2_3_266_FinalData.root
- merged_RS70_LH2_1_264_FinalData.root

2 Kinematic Phase Space

Kinematic Binning Definition:

- **Mass Bins (GeV):** [4.2, 4.5, 4.8, 5.1, 5.4, 5.7, 6.0, 6.3, 6.6, 6.9, 7.5, 8.7]
- p_T **Bins (GeV):** [0.0, 0.32, 0.49, 0.63, 0.77, 0.95, 1.18, 1.8]
- x_F **Bins:** [0.0, 0.85] with a fixed step width of 0.05.

3 Event Selection Criteria

Dimuon Kinematics:

- $4.2 < Mass < 8.8$ GeV, $-0.1 < x_F < 0.95$, $0.05 < x_T \leq 0.58$, and $|\cos\theta| < 0.5$.
- Vertex cuts: $-280 < dz < -5$, target/dump tracking separation criteria met.
- Transverse momentum bounds: $|dp_x| < 1.8$, $|dp_y| < 2.0$, and $dp_x^2 + dp_y^2 < 5$.

Track Quality & Acceptance:

- $\chi^2_{target} < 15$, $\chi^2_{dimuon} < 18$, and $\chi^2/(nHits - 5) < 12$.
- Hit requirements: nHits > 13 per track, $nHits_1 + nHits_2 > 29$, St1 total > 8.
- Track vertex longitudinal limits: $-320 < z_v < -5$.
- A dynamic beam offset correction is applied to y-coordinates: offset is 1.6 for runID ≥ 11000 and 0.4 otherwise.

Detector Occupancy Limits:

- Drift chamber hits: $D1 < 400$, $D2 < 400$, $D3 < 400$, and total $D1 + D2 + D3 < 1000$.

4 Plots and Distributions

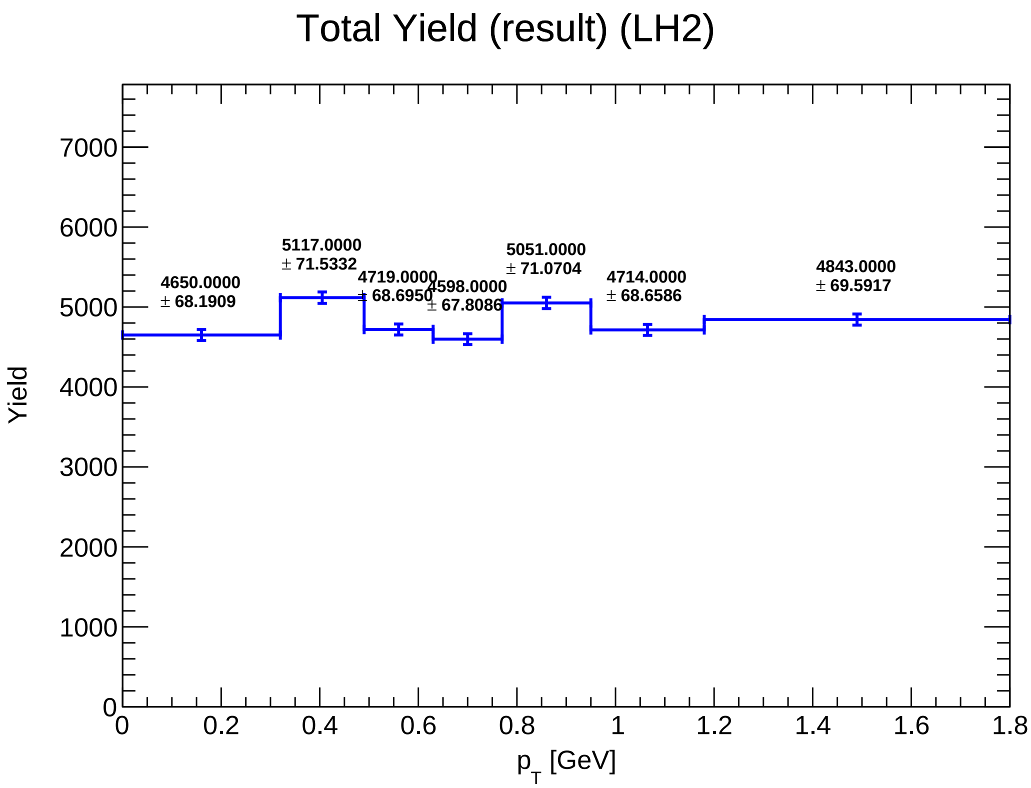

Figure 1: Y_total_LH2

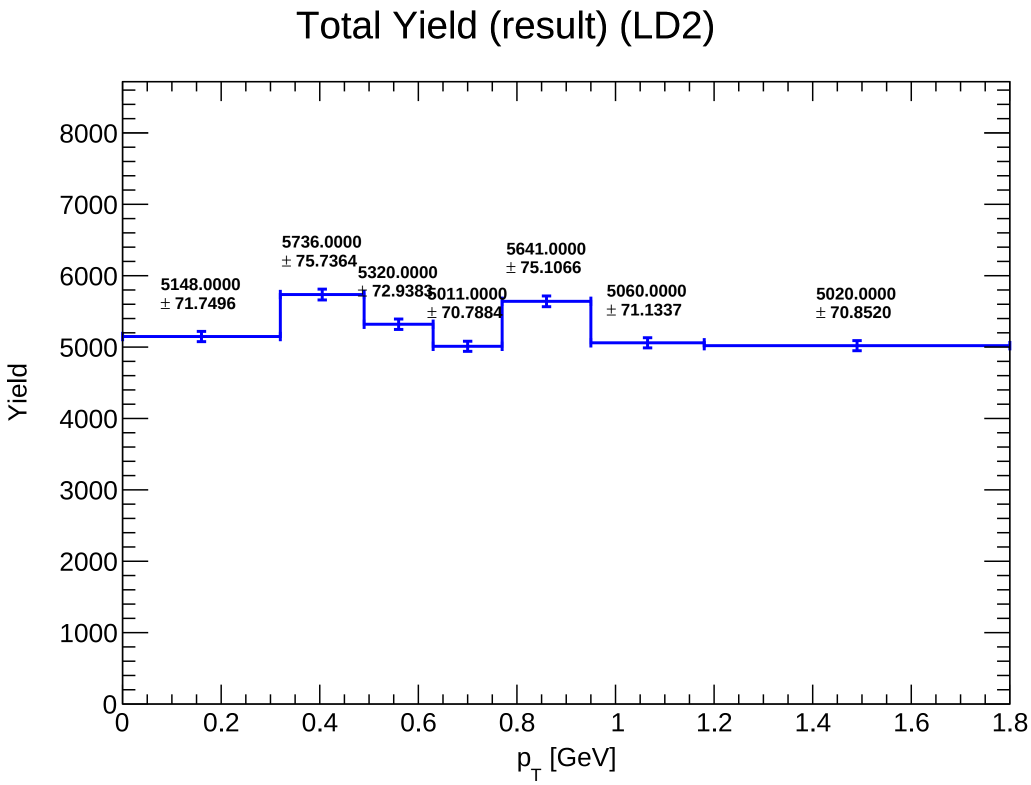

Figure 2: Y_total_LD2

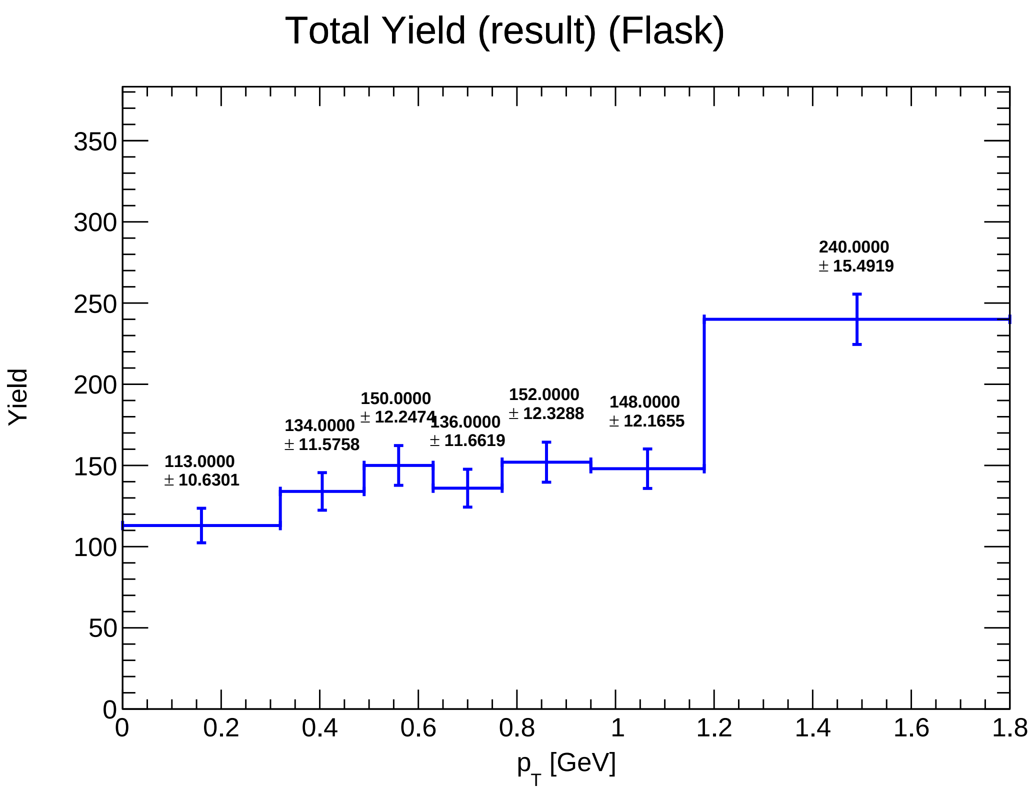

Figure 3: Y_total_Flask

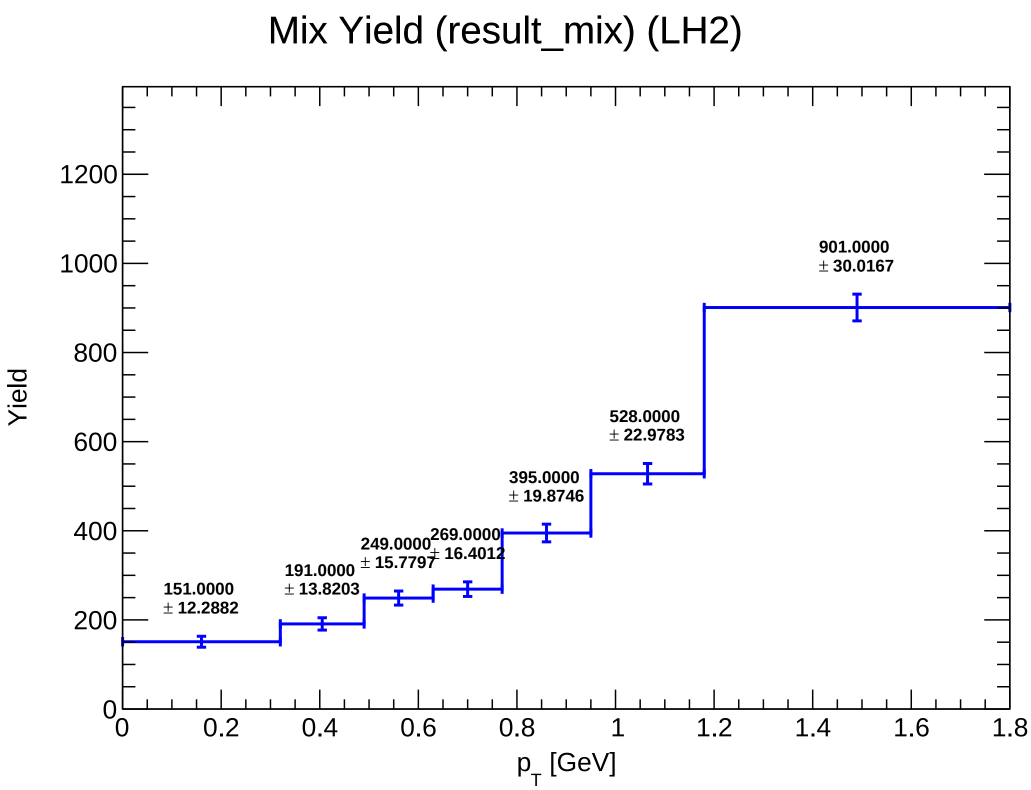

Figure 4: Y_mix_LH2

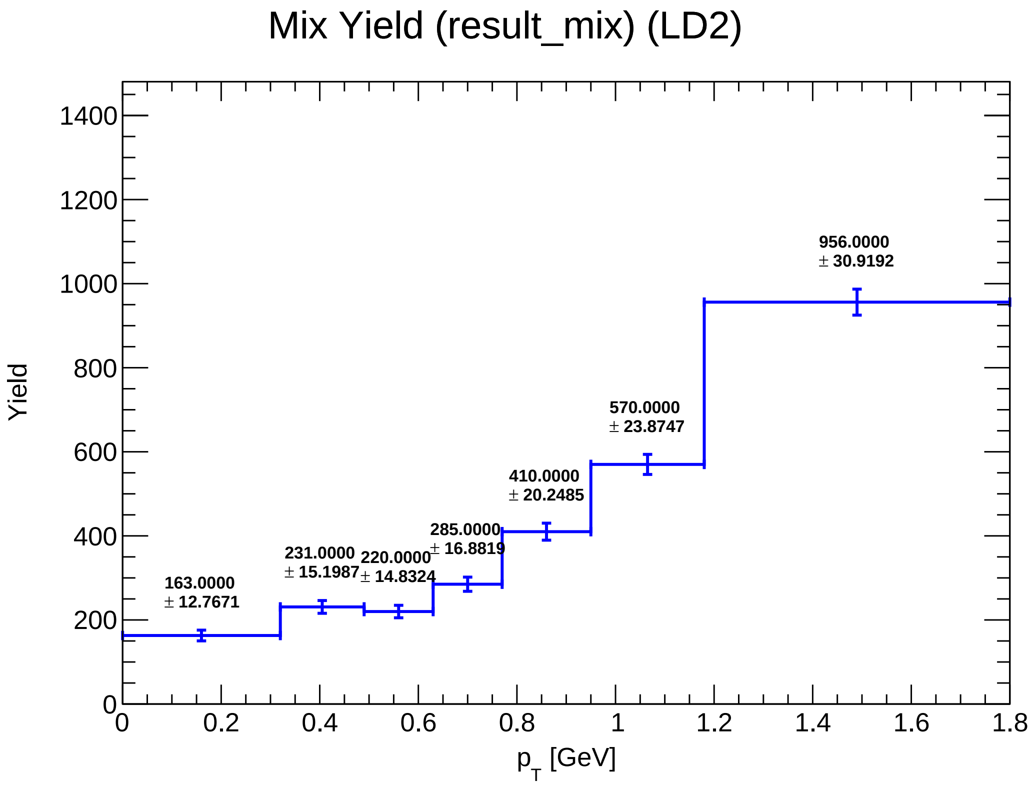

Figure 5: Y_mix_LD2

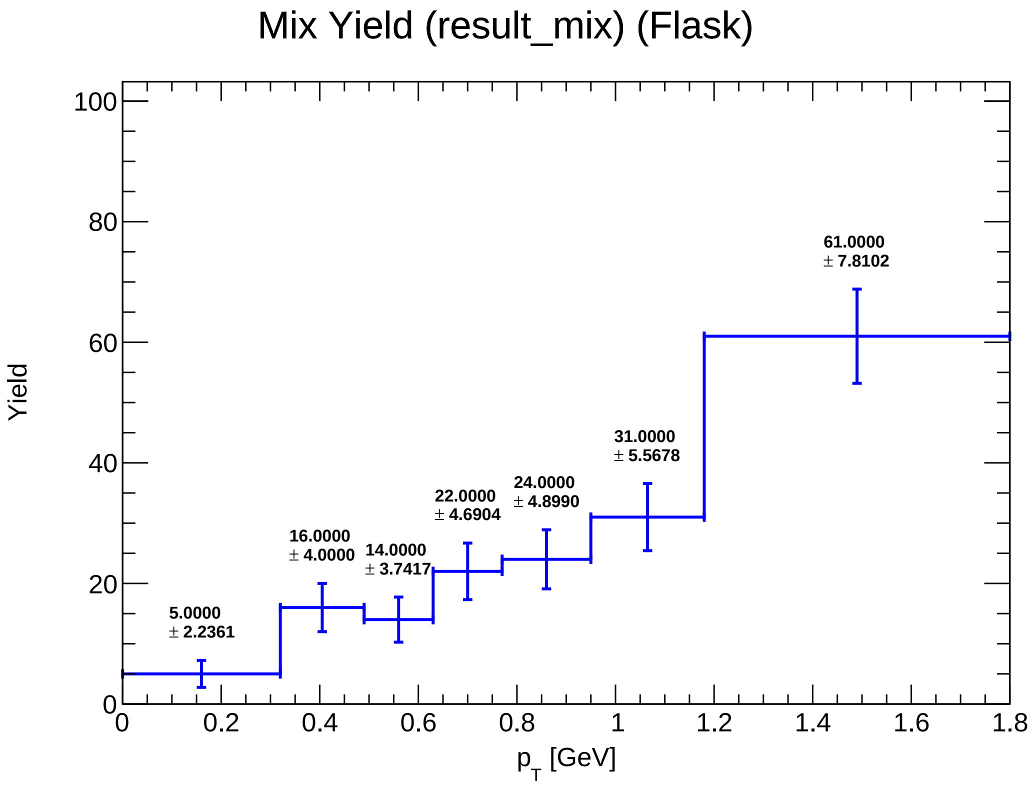

Figure 6: Y_mix_Flask

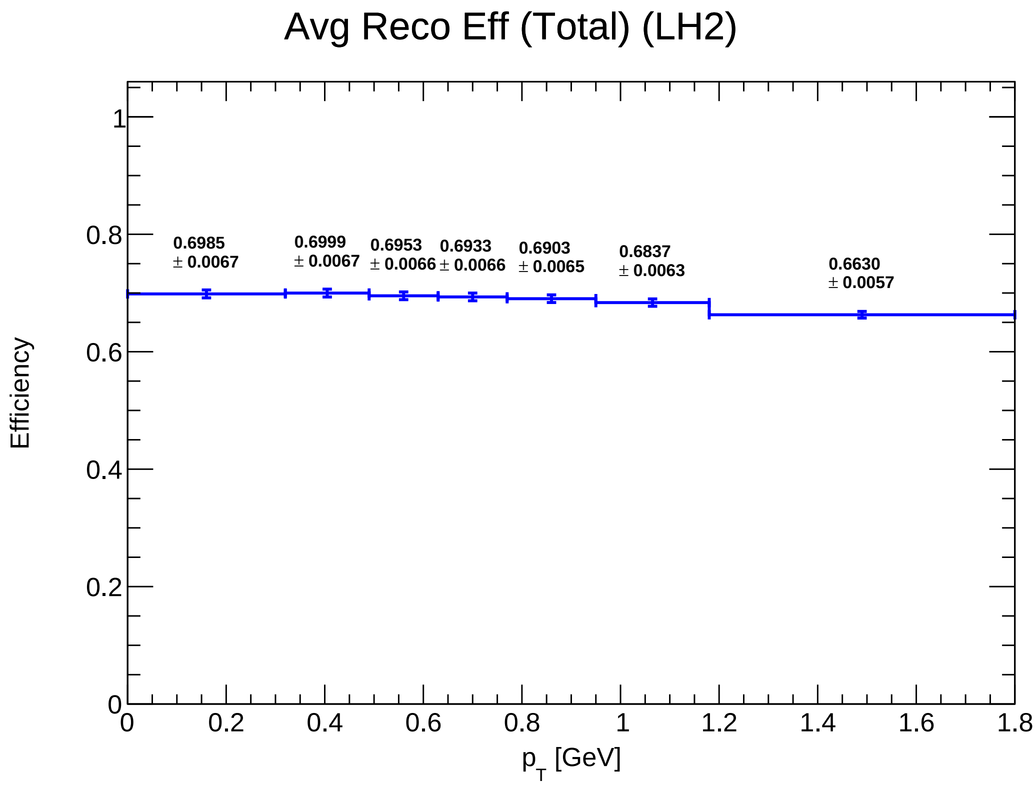

Figure 7: E_total_reco_LH2

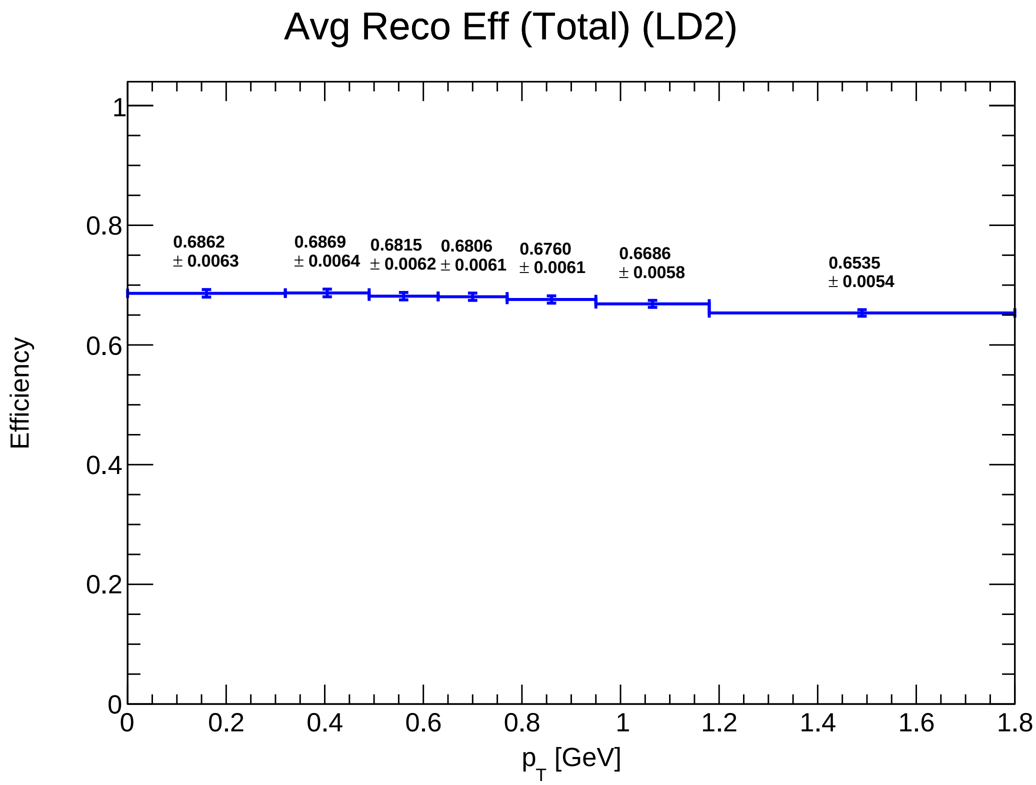

Figure 8: E_total_reco_LD2

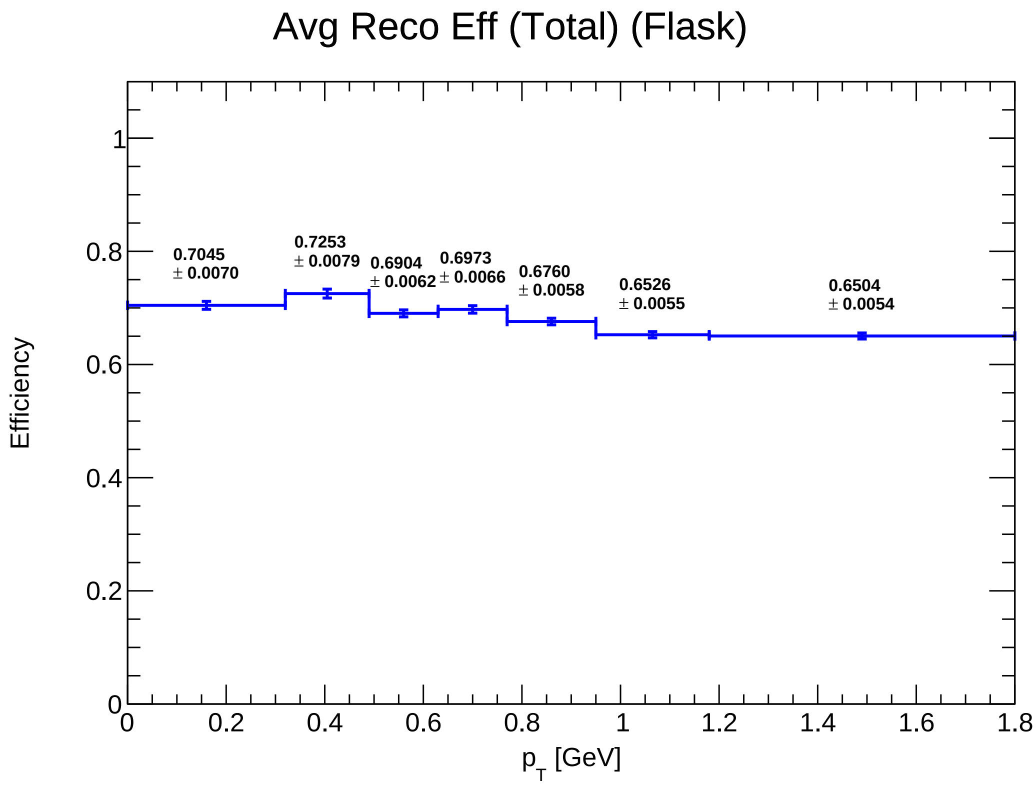

Figure 9: E_total_reco_Flask

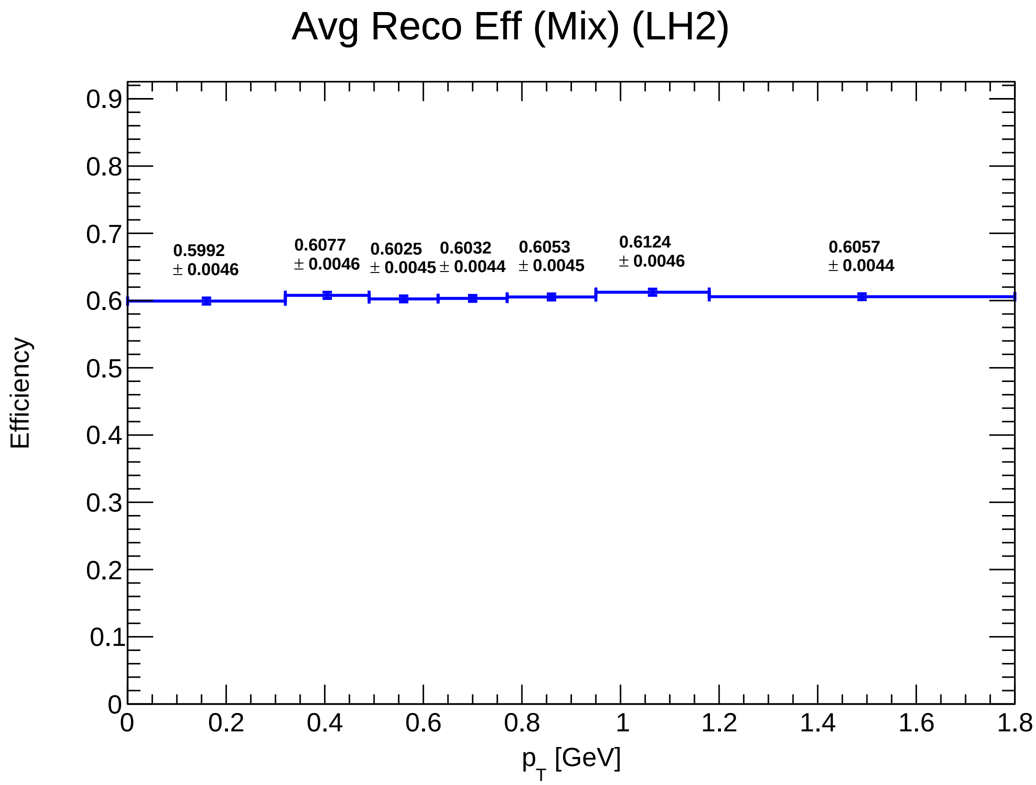

Figure 10: E_mix_reco_LH2

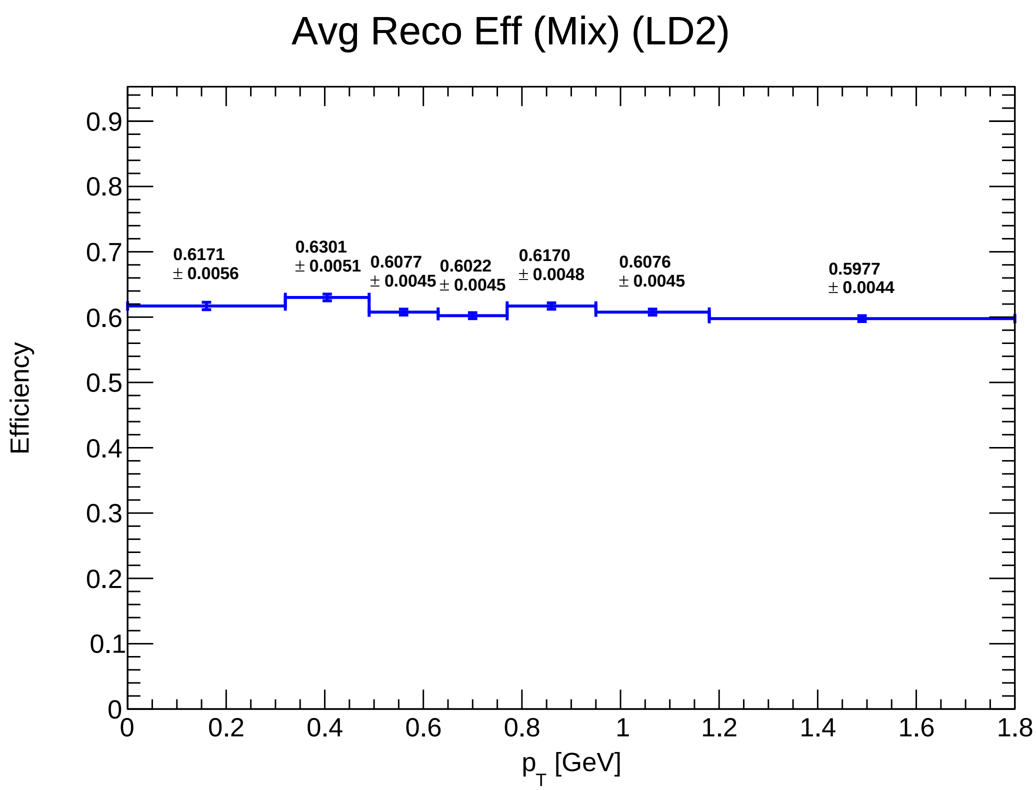

Figure 11: E_mix_reco_LD2

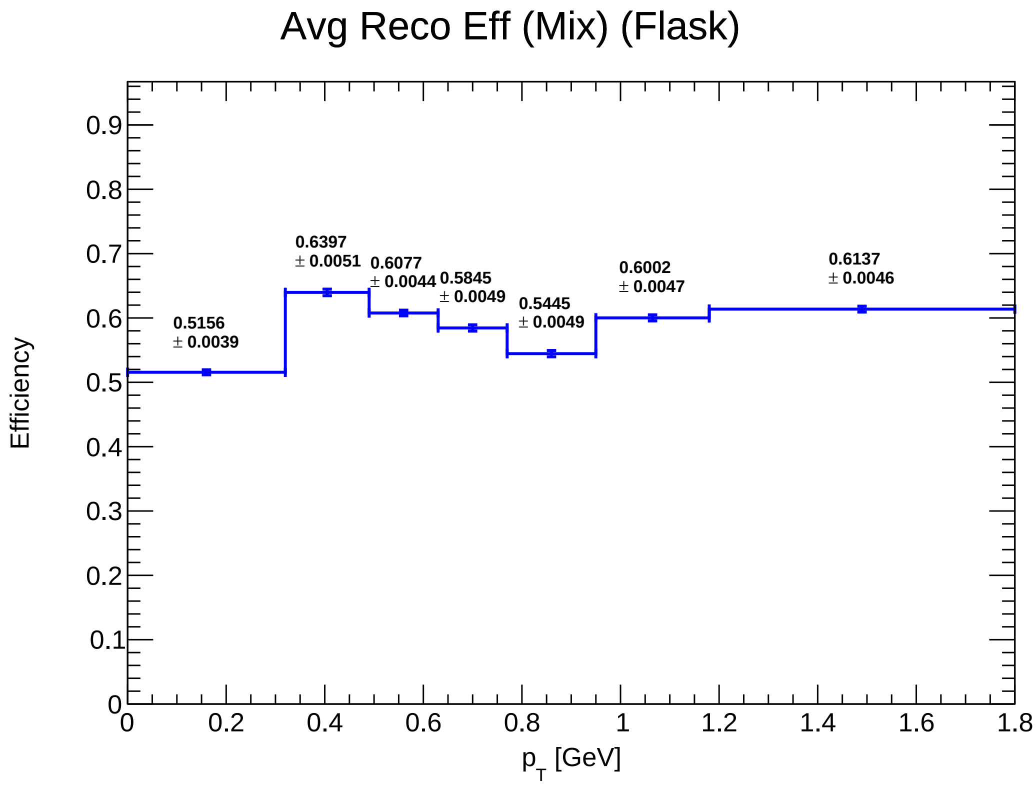

Figure 12: E_mix_reco_Flask

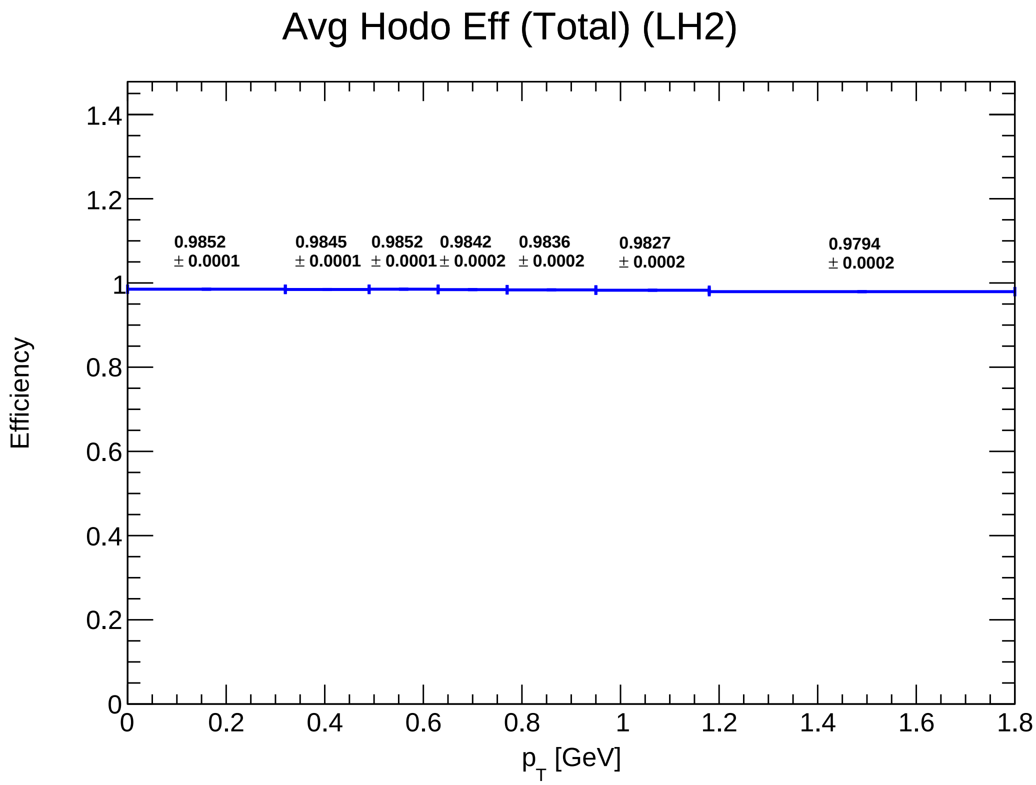

Figure 13: E_total_hodo_LH2

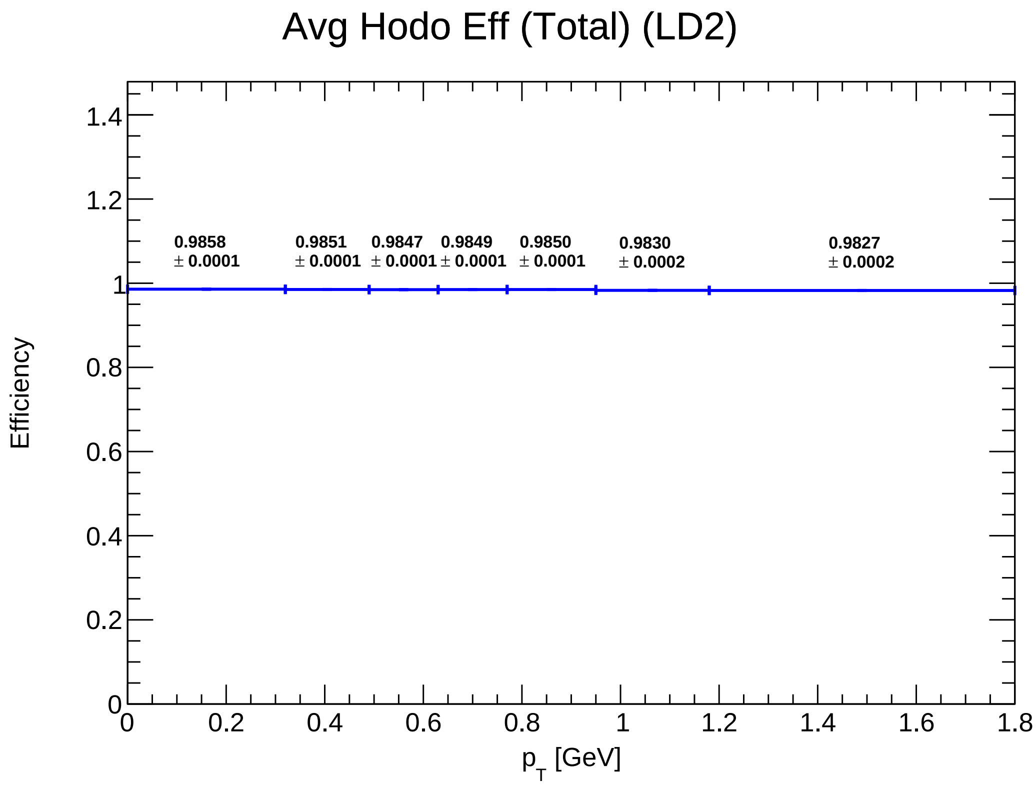

Figure 14: E_total_hodo_LD2

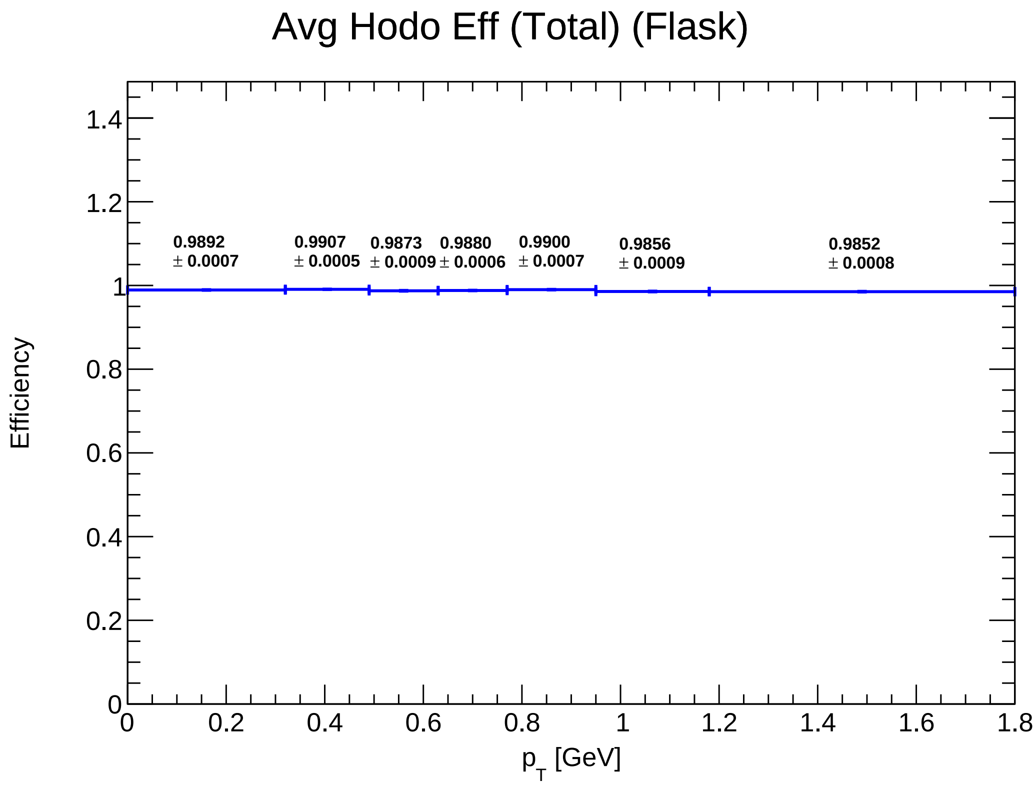

Figure 15: E_total_hodo_Flask

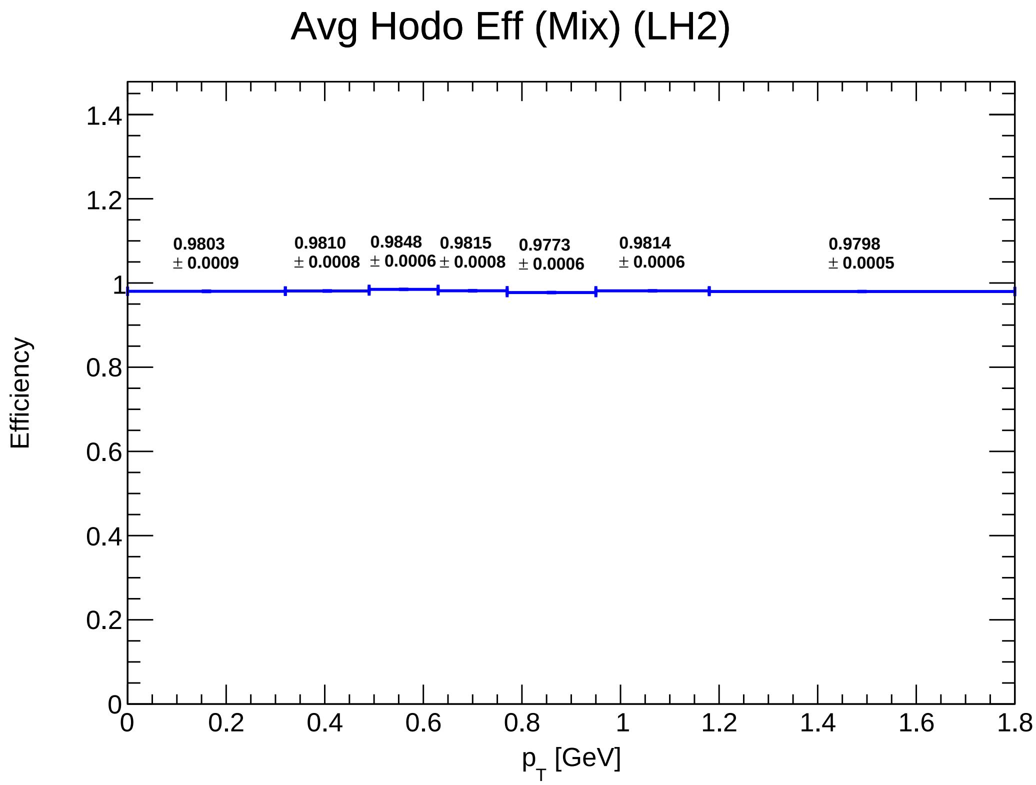

Figure 16: E_mix_hodo_LH2

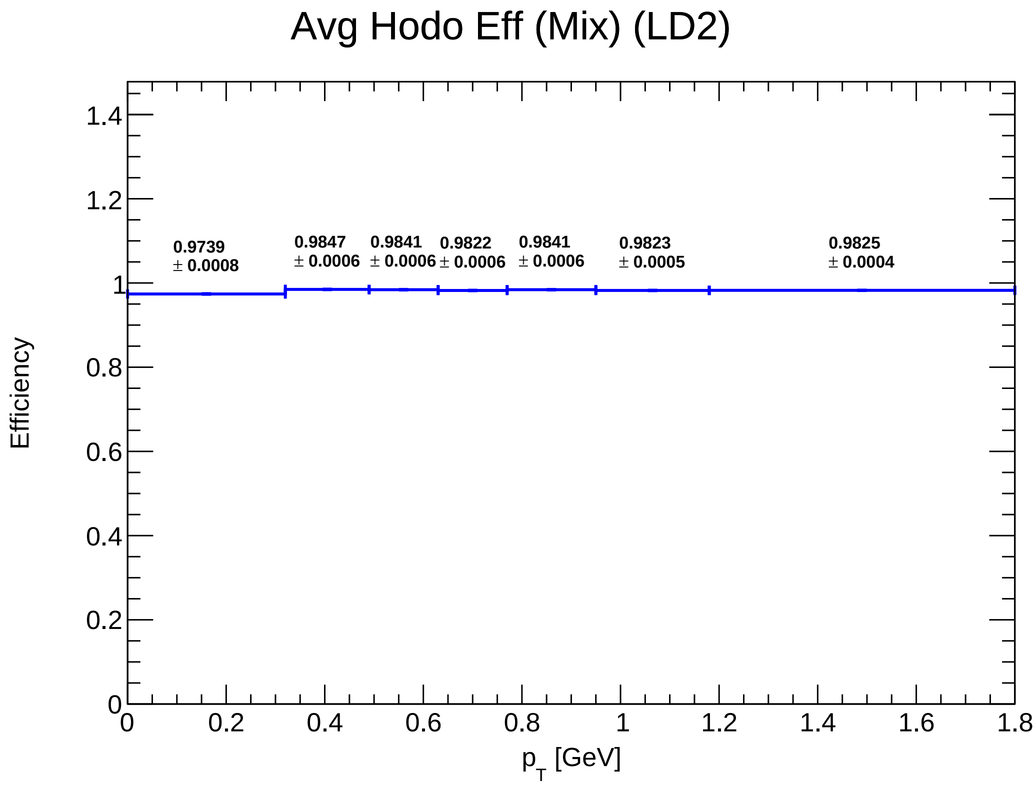

Figure 17: E_mix_hodo_LD2

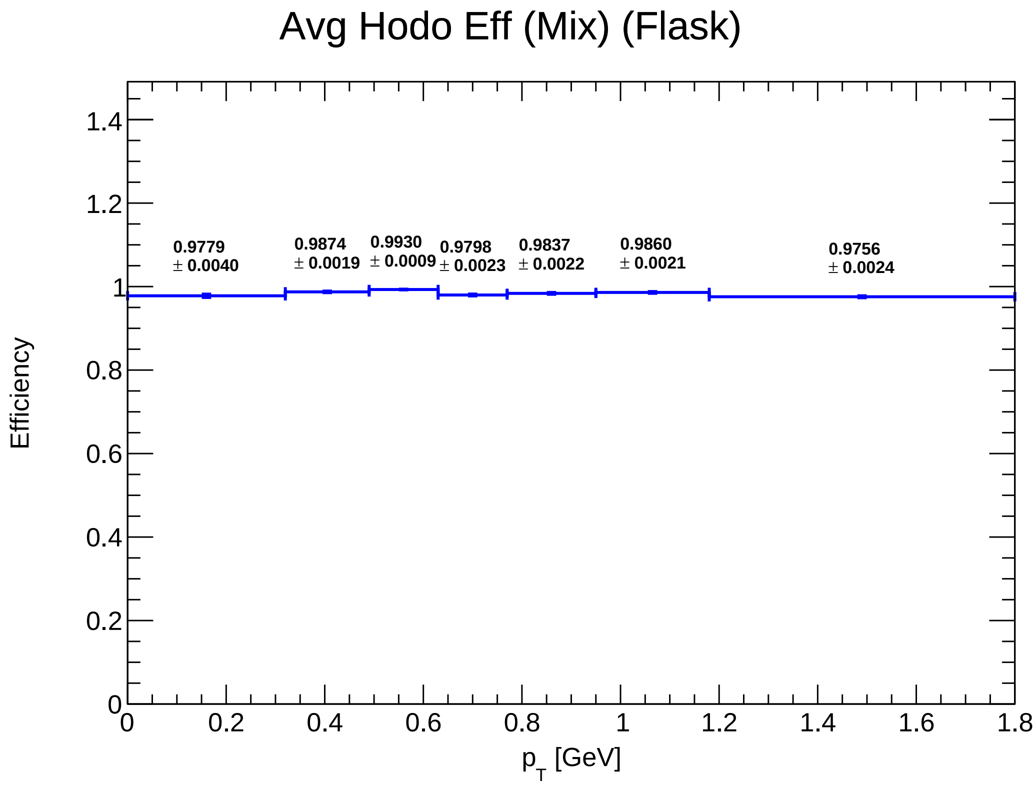

Figure 18: E_mix_hodo_Flask

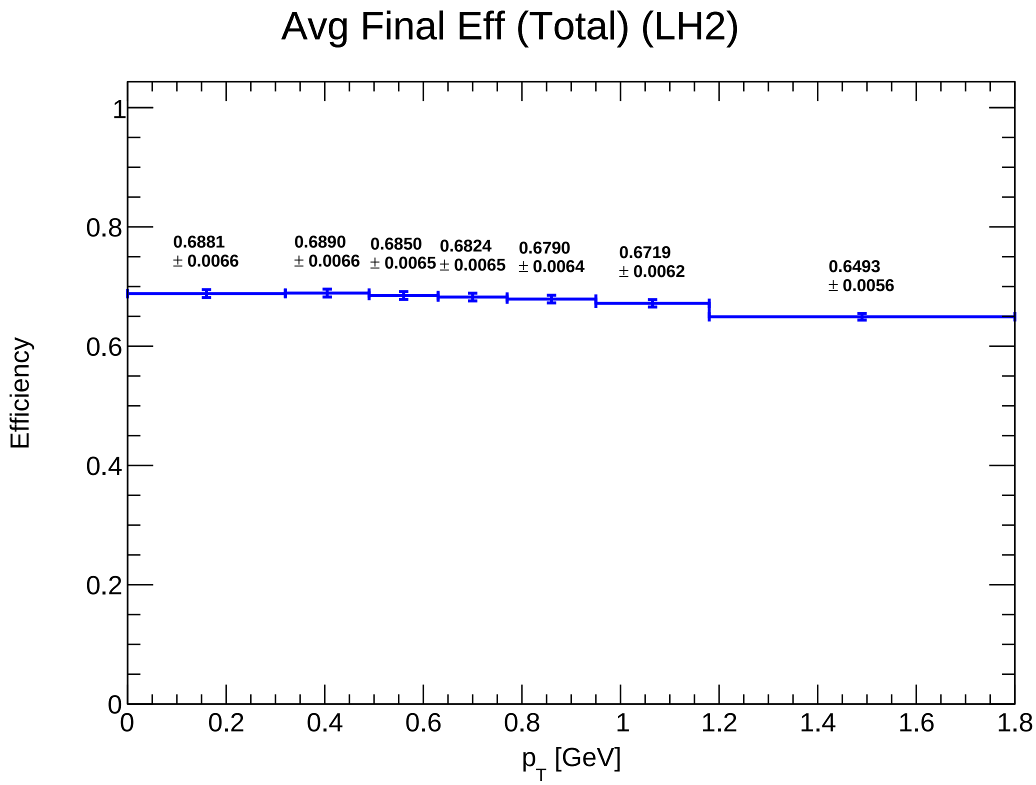

Figure 19: E_total_final_LH2

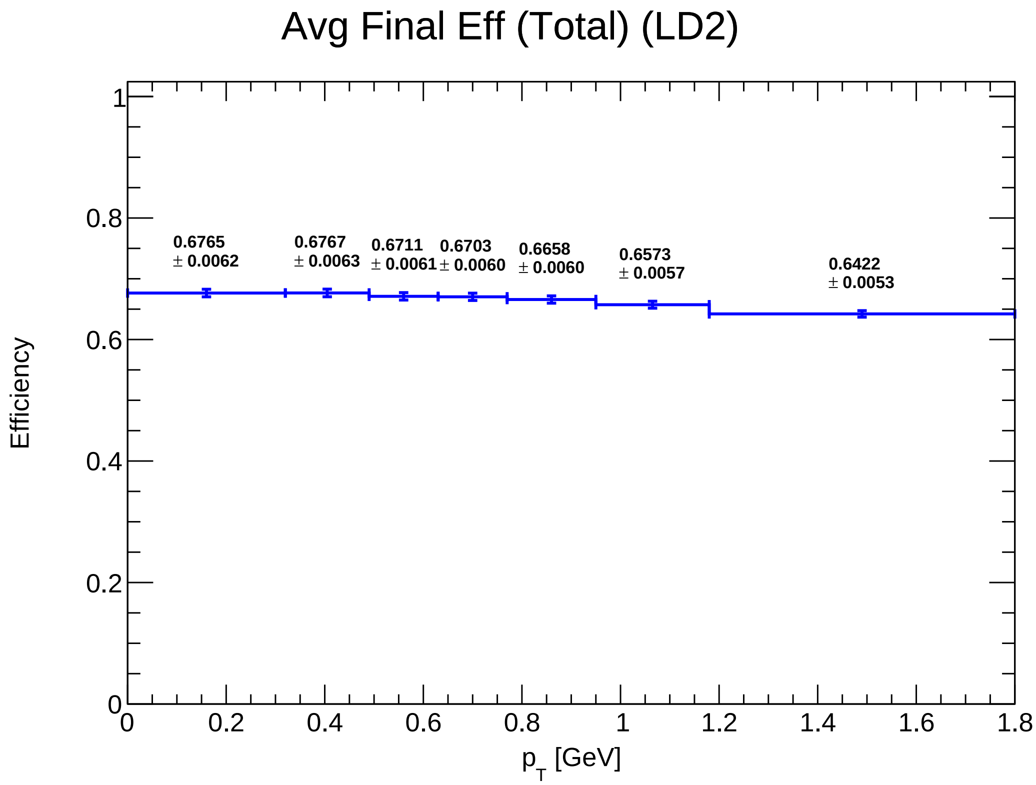

Figure 20: E_total_final_LD2

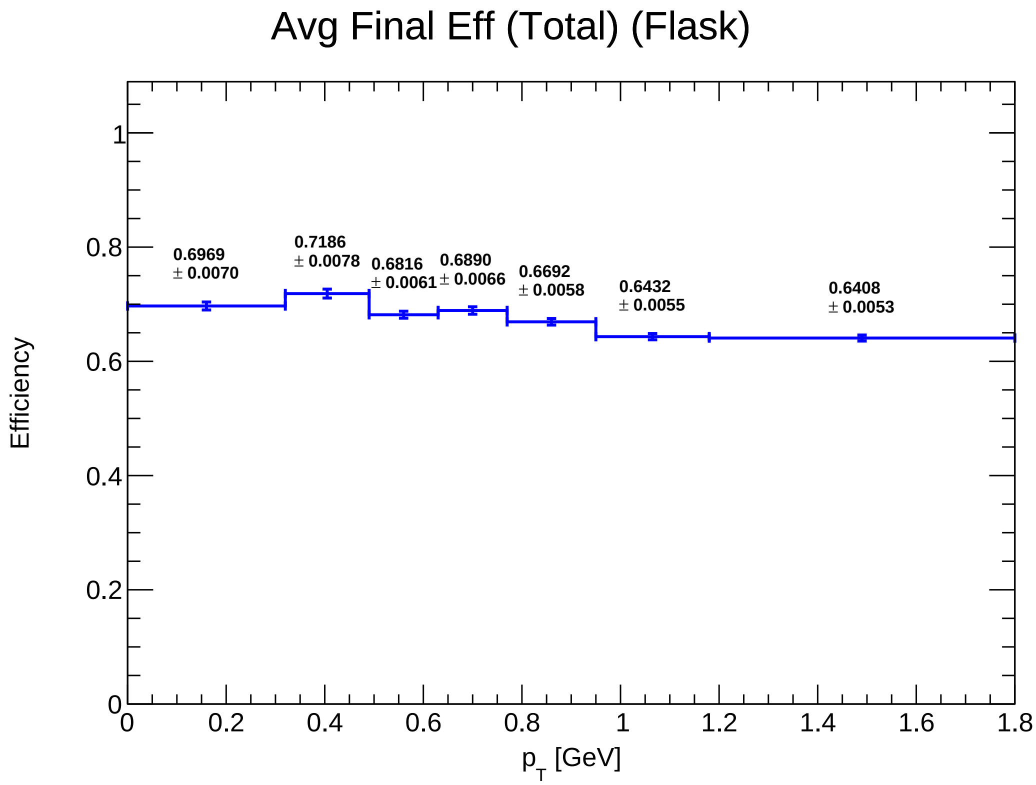

Figure 21: E_total_final_Flask

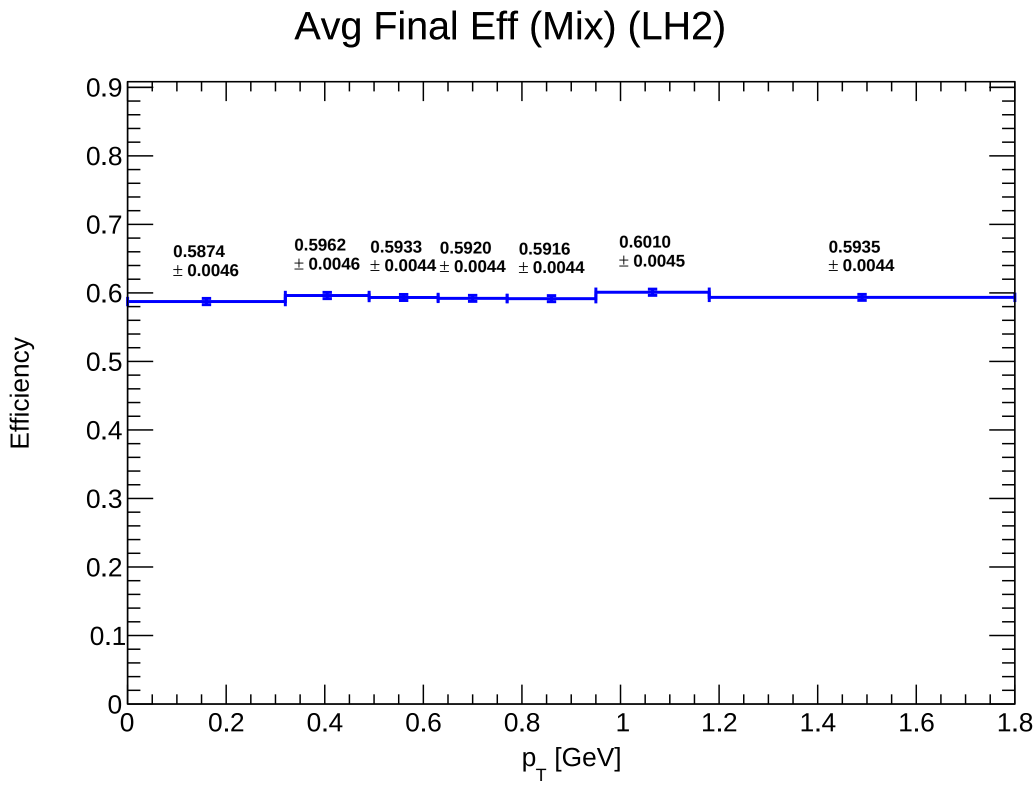

Figure 22: E_mix_final_LH2

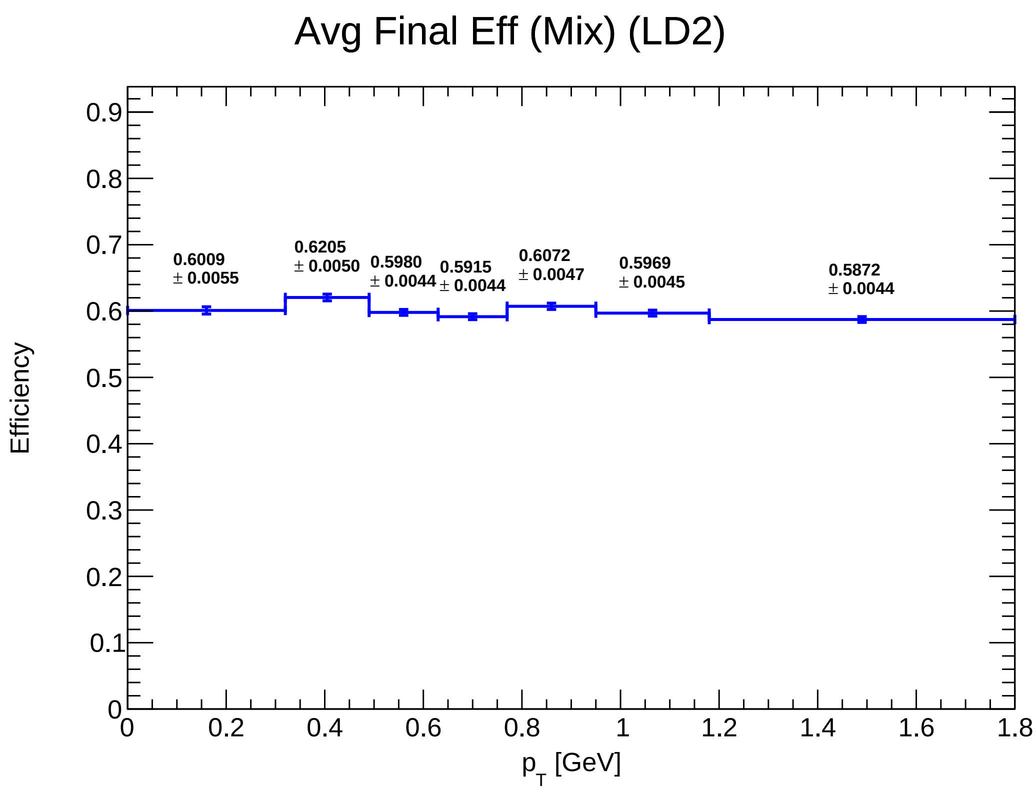

Figure 23: E_mix_final_LD2

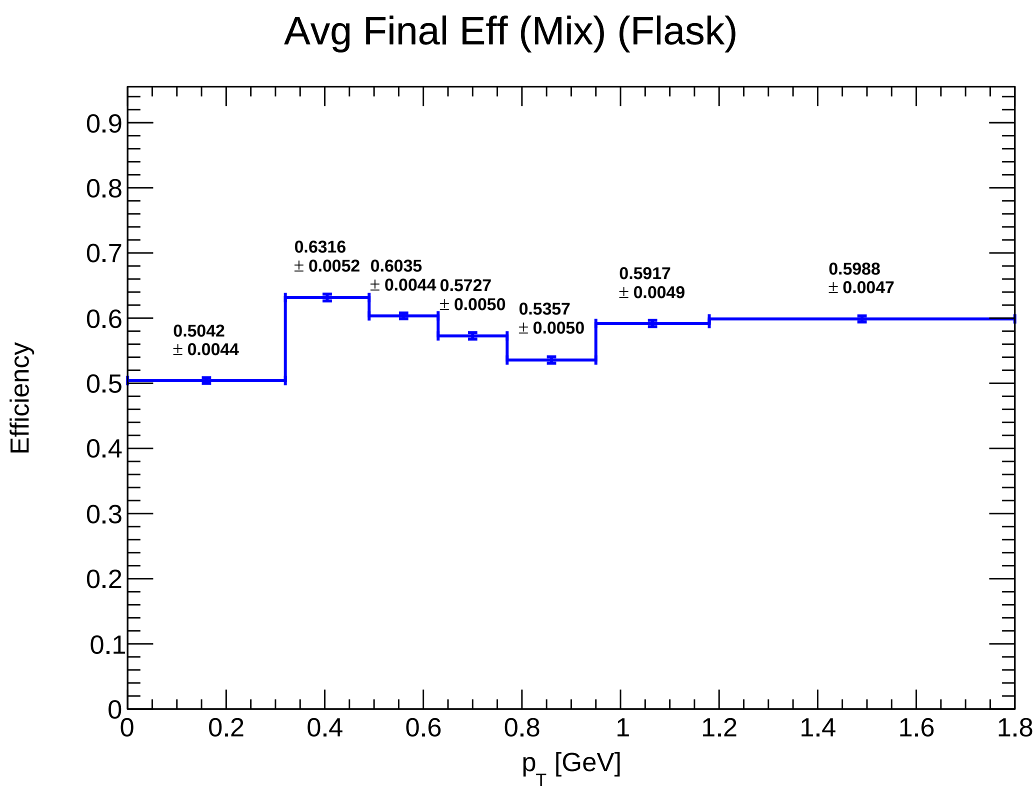

Figure 24: E_mix_final_Flask

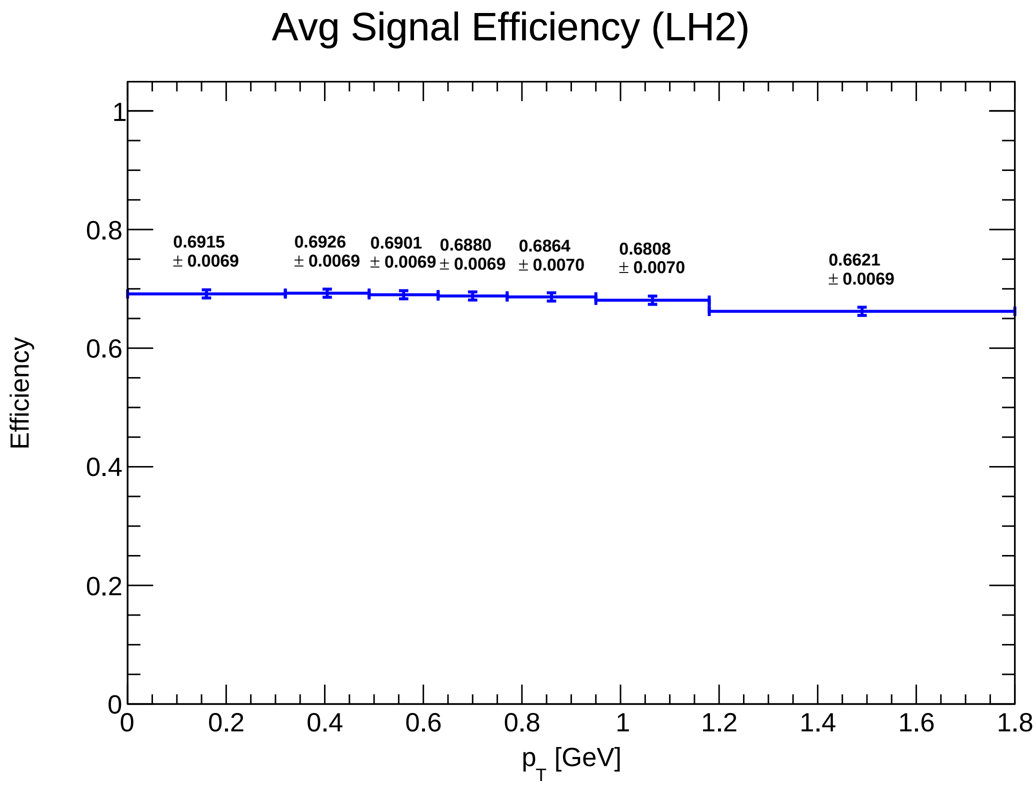

Figure 25: E_final_signal_LH2

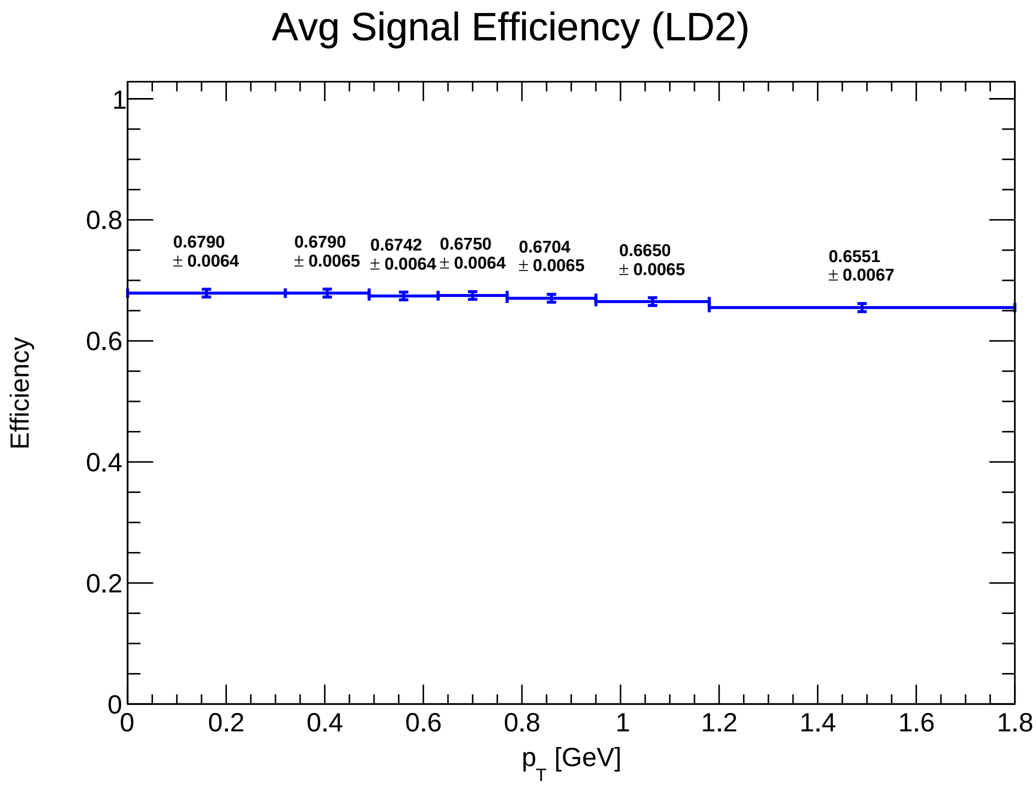

Figure 26: E_final_signal_LD2

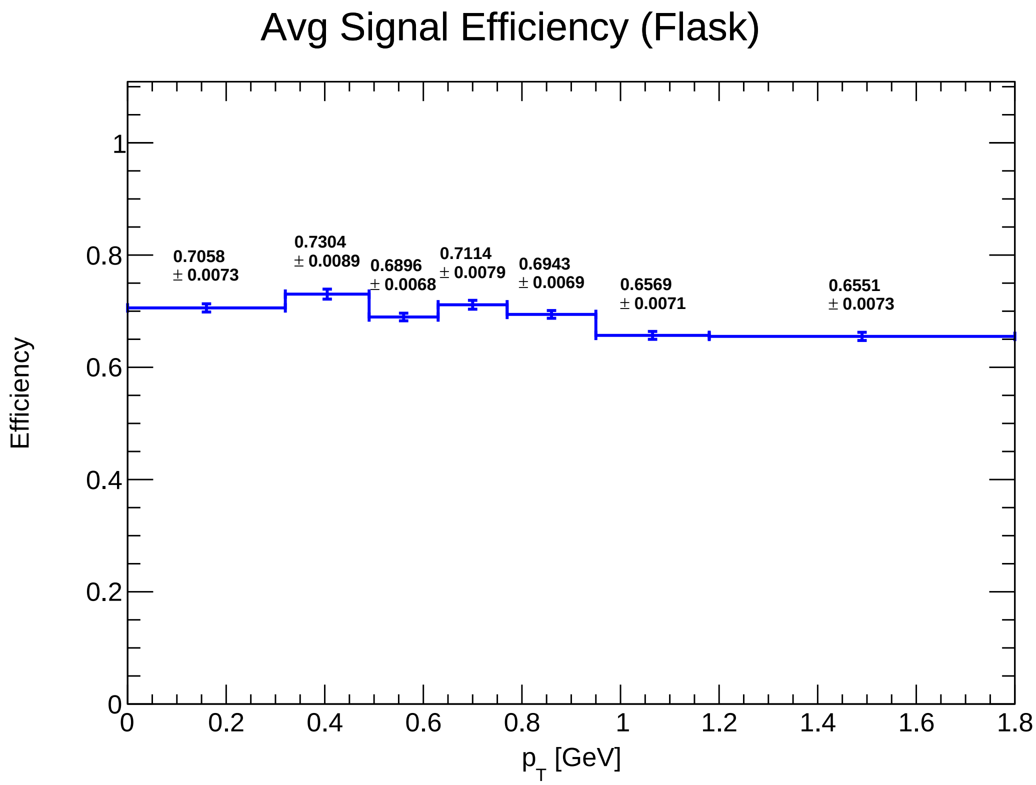

Figure 27: E_final_signal_Flask

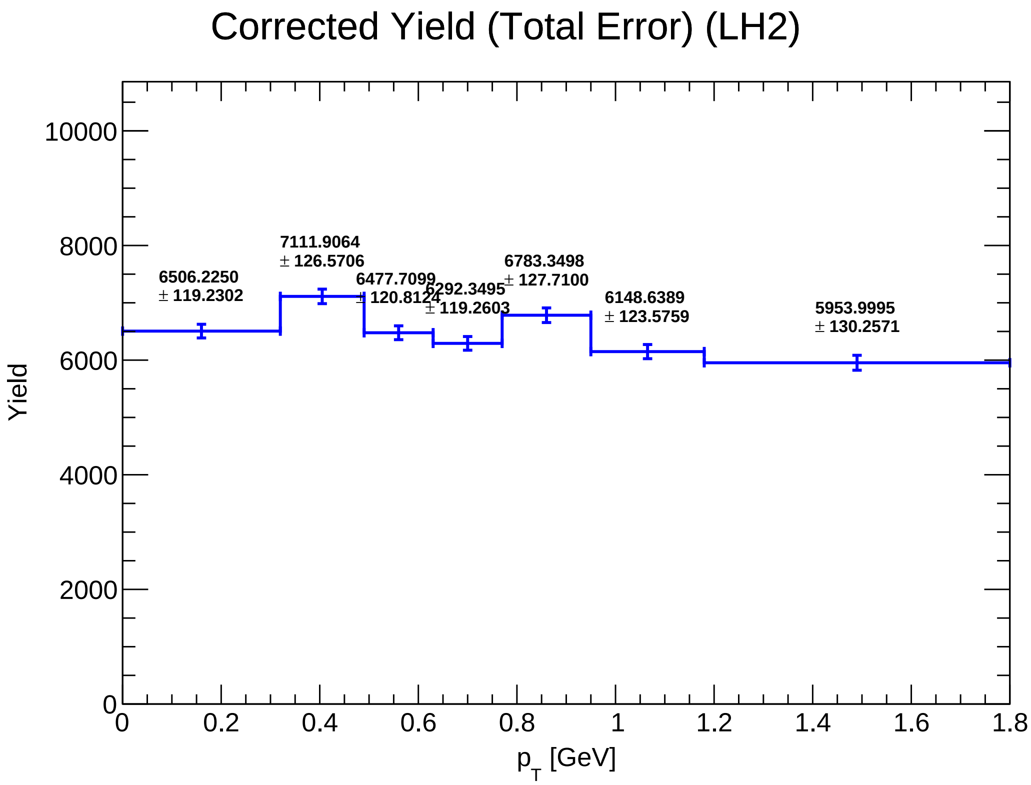

Figure 28: Y_corrected_LH2

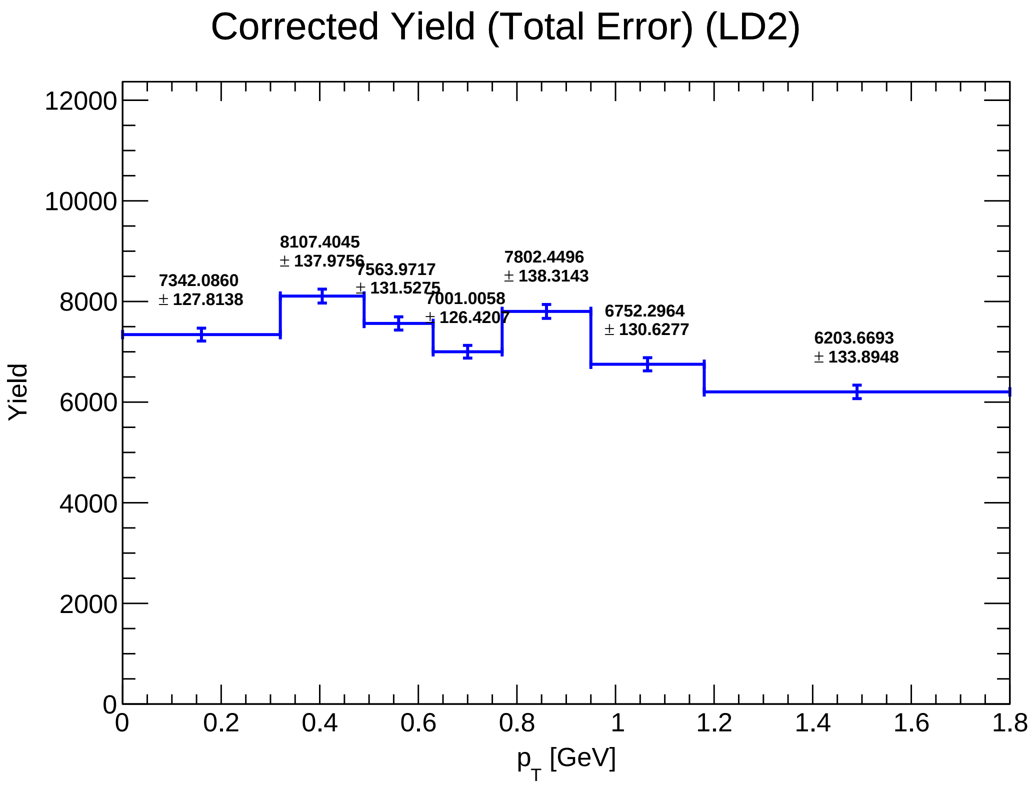

Figure 29: Y_corrected_LD2

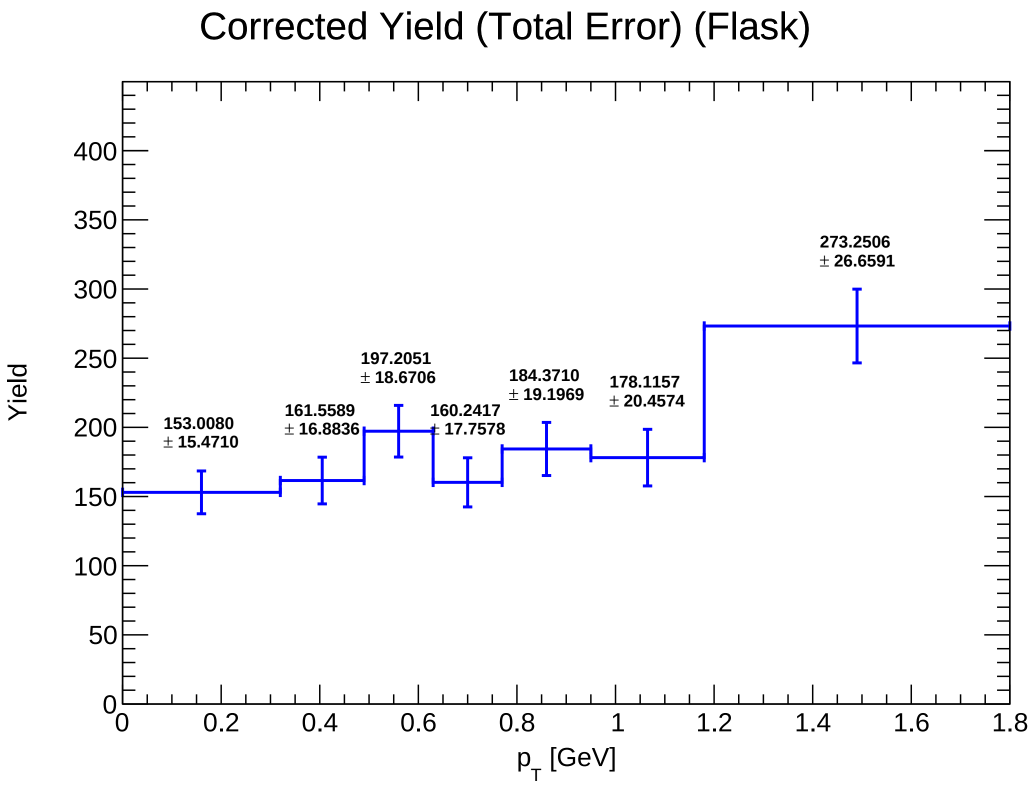

Figure 30: Y_corrected_Flask

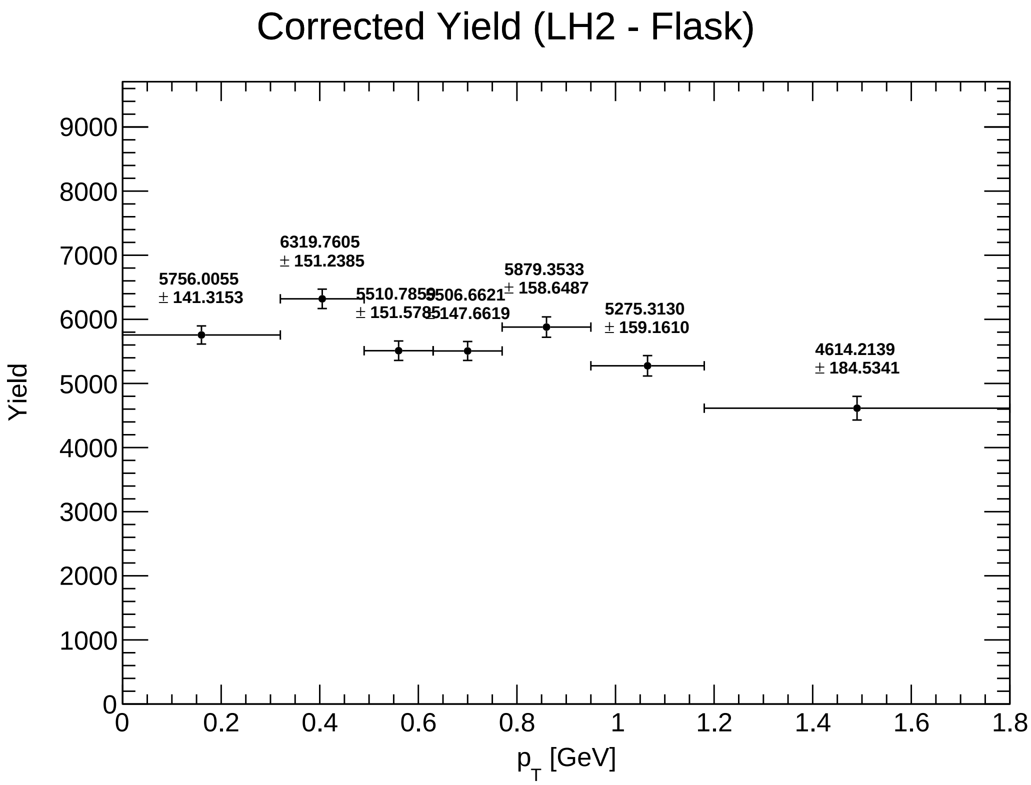

Figure 31: Y_corrected_Subtracted_LH2

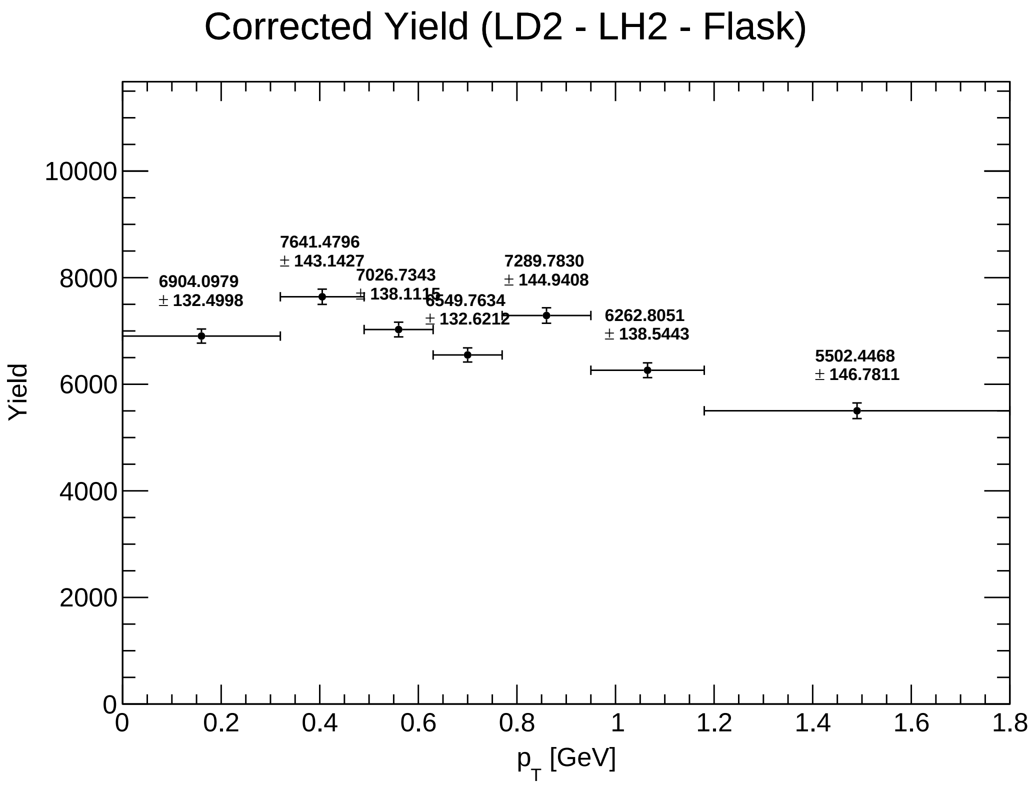

Figure 32: Y_corrected_Subtracted_LD2

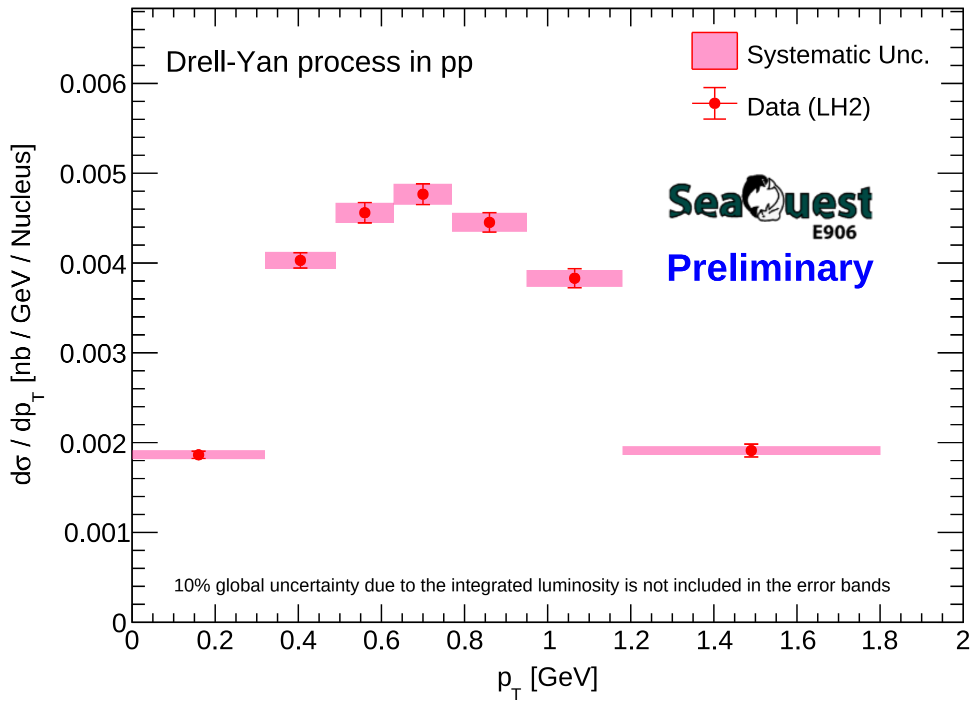

Figure 33: CrossSection_LH2_geom_vs_pT_with_logo

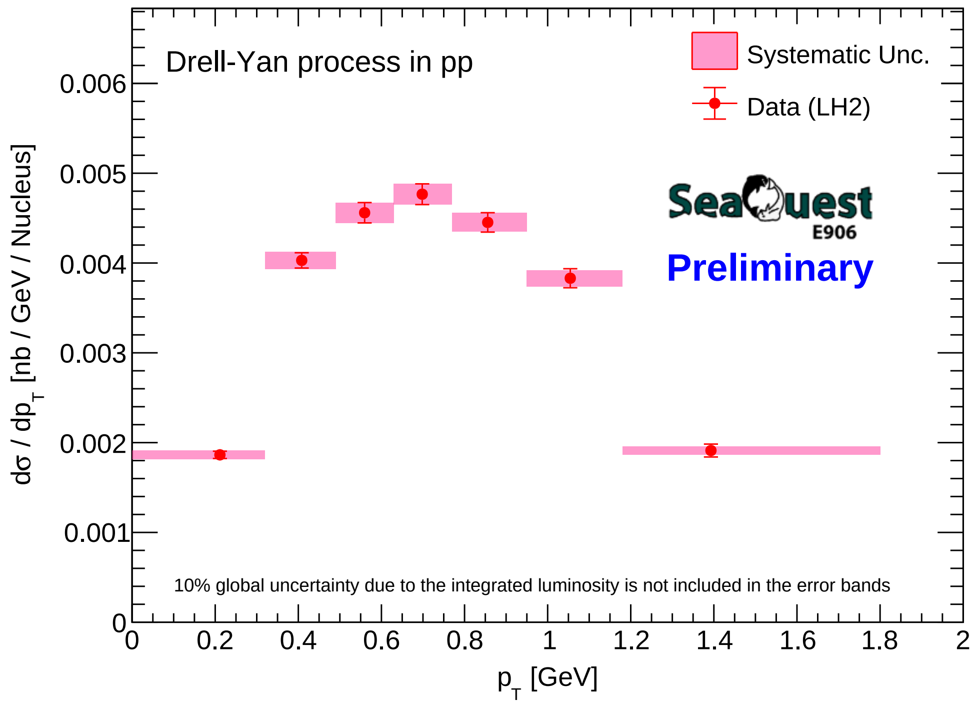

Figure 34: CrossSection_LH2_true_pt_vs_pT_with_logo

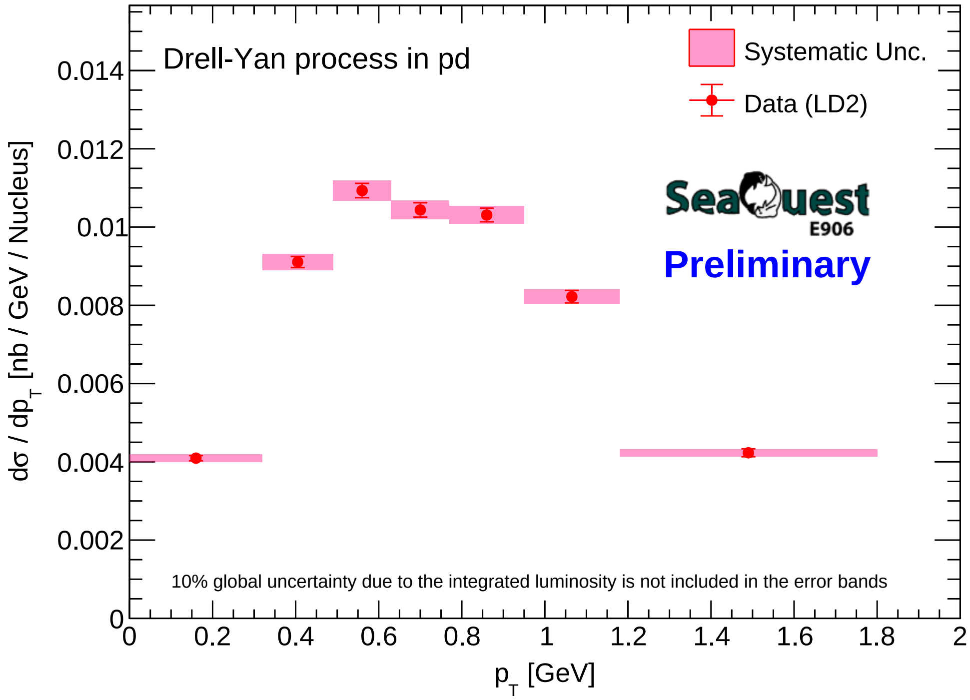

Figure 35: CrossSection_LD2_geom_vs_pT_with_logo

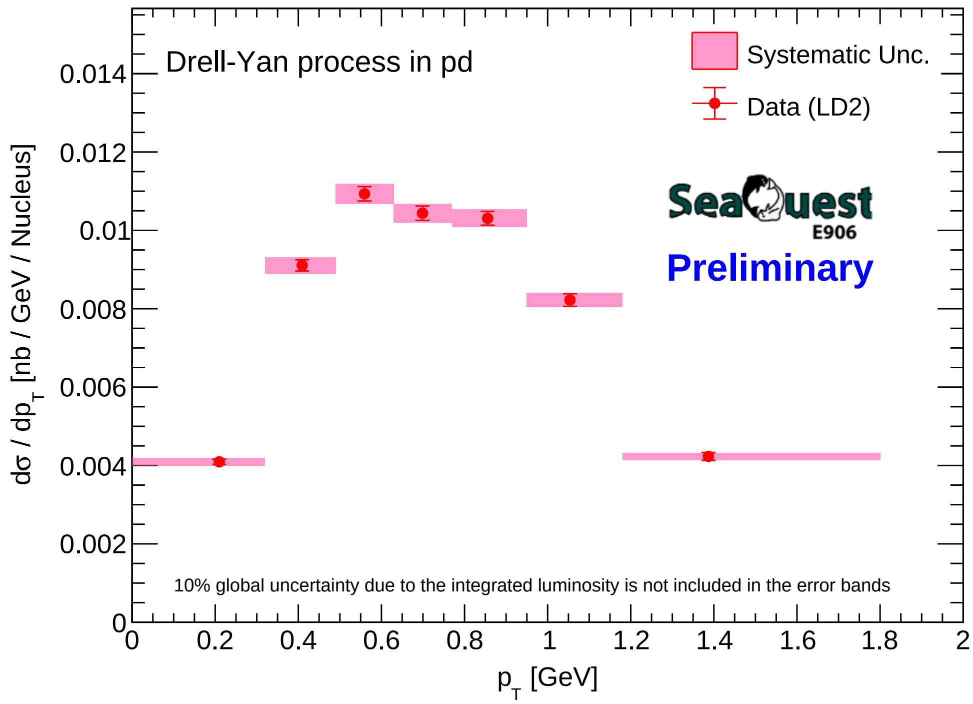

Figure 36: CrossSection_LD2_true_pt_vs_pT_with_logo

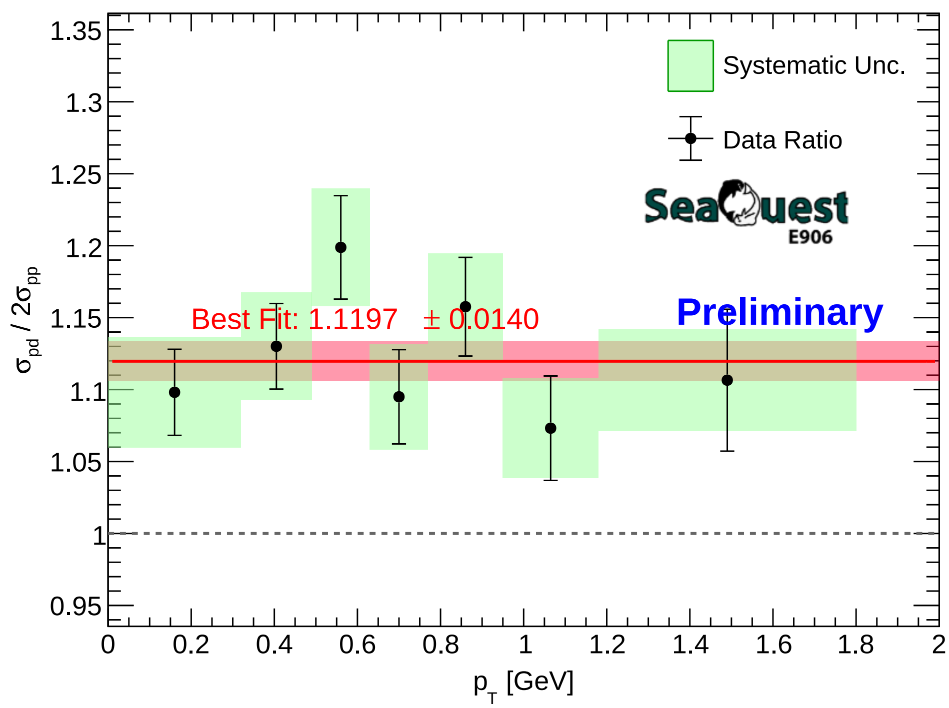

Figure 37: CrossSection_Ratio_pd_2pp_vs_pT_geom_with_logo

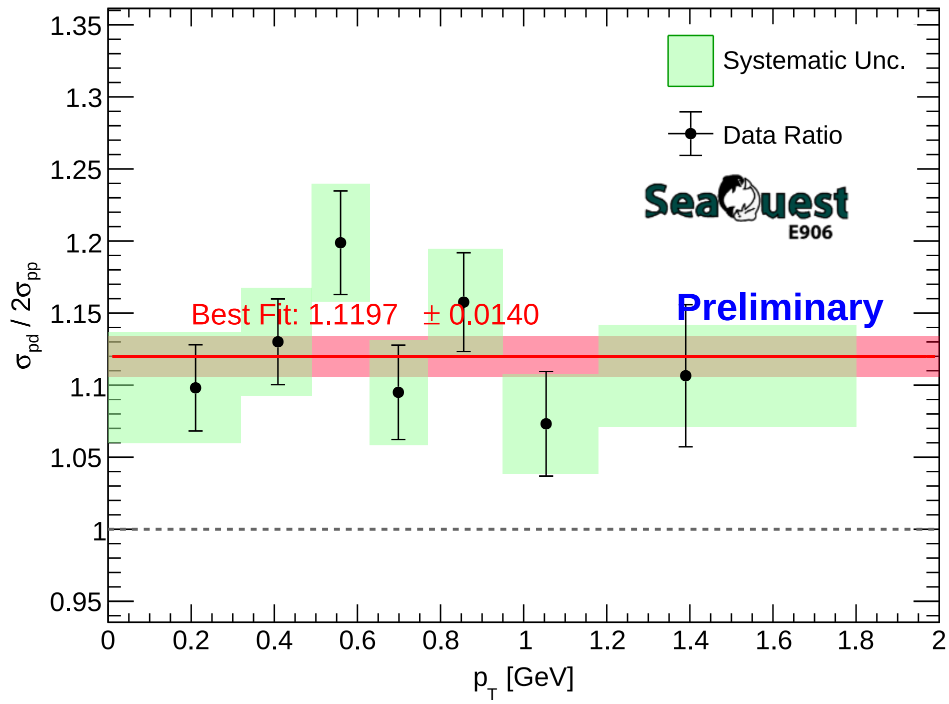

Figure 38: CrossSection_Ratio_pd_2pp_vs_pT_true_pt_with_logo

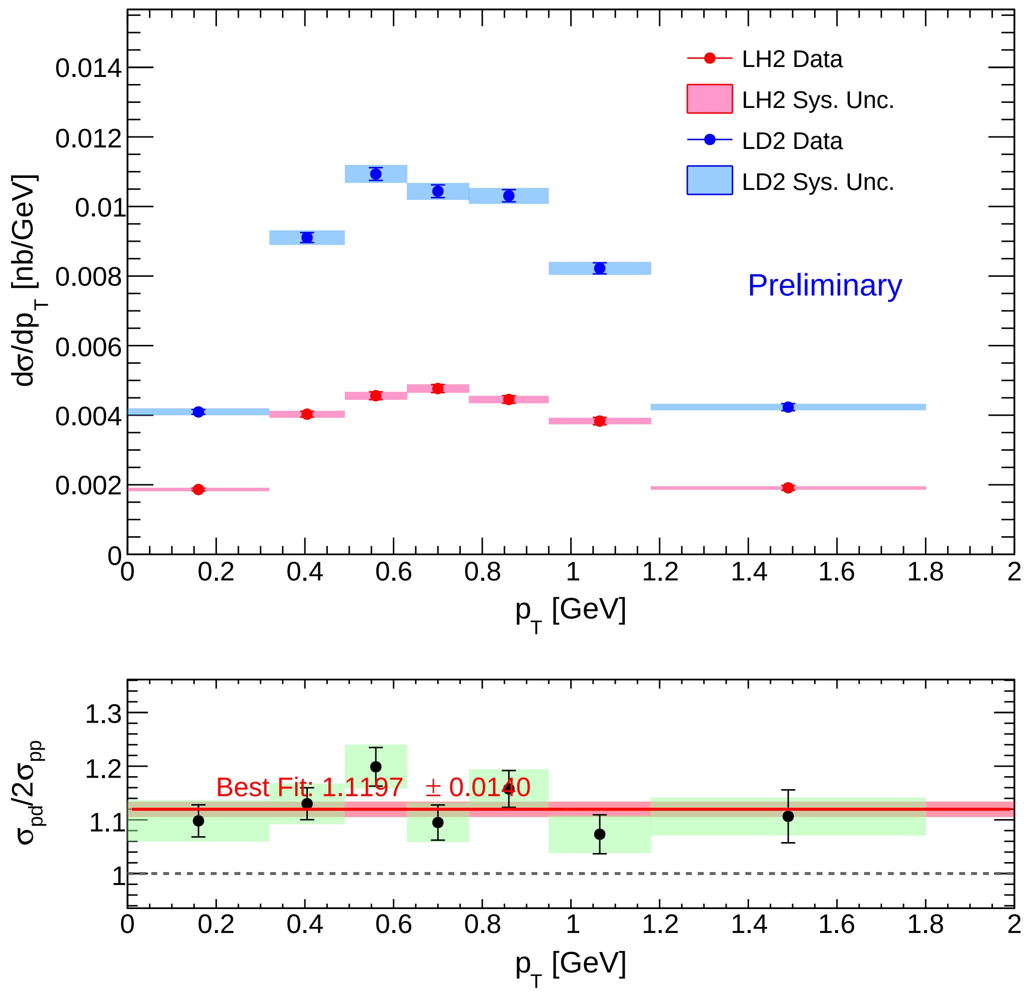

Figure 39: Combined_XSec_Ratio_vs_pT_geom

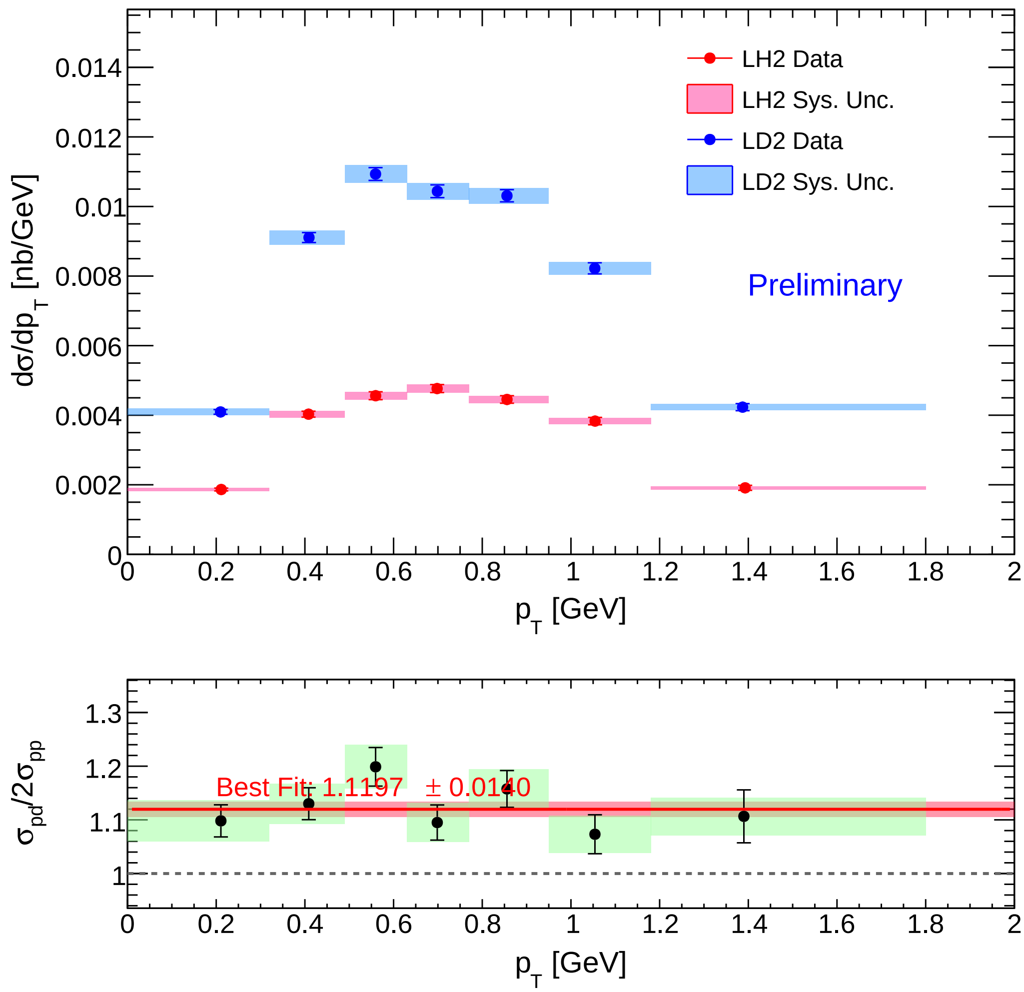

Figure 40: Combined_XSec_Ratio_vs_pT_true_pt